

JKBOSE 2026

CLASS 11TH

NEET = 1-2 Questions
Max.

THE LIVING WORLD

LECTURE- 02

BOTANY

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Topics To Covered

- 1** What is Living?
→ NOT in NCERT (NOT also in BOARD).
→ included in NEET syllabus
- 2** Biodiversity
chat off → Thumbs up
chat on → Thumbs down
- 3** Need of Classification

WHAT IS LIVING?

↳ Five criteria — used to describe living organism:

a. Growth.

b. Reproduction

c. Metabolism

d. Cellular organisation

e. Consciousness

not defining property.

are defining properties

* Defining property: property which is present in all the living organisms without any exception.

Ex. Metabolism, Cellular organisation, Consciousness.

* Characteristic property: property which is not in living organisms
→ It can also be present in non-livings.

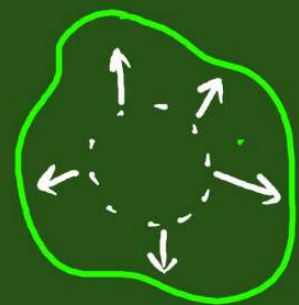
Ex. Growth, Reproduction.

(a) GROWTH

↳ It is permanent, irreversible increase in

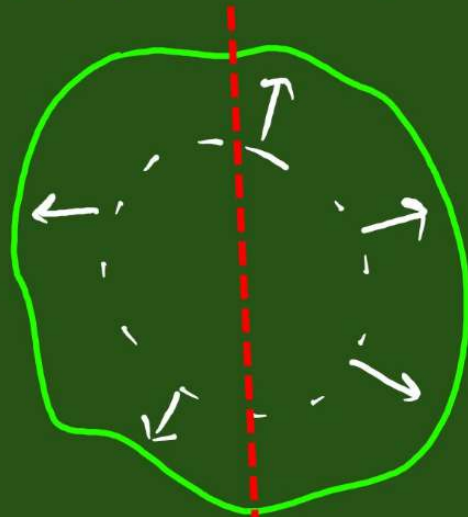
Size of cell
mass of cell
height
No. of cells.

↳ Growth involves $\left\{ \begin{array}{l} \text{increase in Size/mass} \\ \text{increase in number} \end{array} \right\}$ Twin Characters of Growth



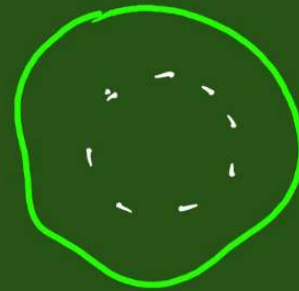
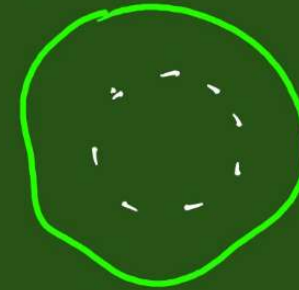
Amoeba
(unicellular organism)

Increase in
Size/mass



Binary
fission

Increase in
no.



Two daughter Amoeba.

- * Plants — unlimited growth / Continuous growth / indeterminate growth
 ↳ Continuously grow throughout life because of presence of meristematic tissue
- * Animals — limited growth / determinate growth.
 ↳ show growth upto a certain stage of life. (No meristematic tissue).

NOTE: Growth is not a defining property.

Reason: = Growth is int in all living organisms. { Due to increase in protoplasm int inside the cell }
 i.e, why growth in livings is - Internal.
 (protoplasmic growth).

= Non-livings also show growth
 due to accumulation of dust on mountains, sand-dunes etc
 i.e, why growth in non-livings is - External / Extrinsic

Q. Internal / intrinsic growth is a defining property?

Ans. YES ($\because$ Internal growth is +nt in all living organisms
and -nt in non-livings)

imp.

Note:-

Growth in livings, involves

- Synthesis of living material like pm, cytoplasm etc.
- Synthesis of dead material like cell wall (plants)

(b) REPRODUCTION

↳ It is the production of new individuals. (Biological process).

↳ Two types: a. Sexual reproduction:

- preferred by higher organisms.
- involves — gametogenesis & fertilization.
- Slow process.

b. Asexual reproduction: preferred by lower organisms.

- No gametogenesis, NO fertilization.
- Rapid process.

Various types.

Binary fission

Ex. Unicellular forms
like Amoeba,
Paramecium

Budding

Ex. yeast,
Hydra

Fragmentation

Ex. Spirogyra, Ulothrix,
Rhizopus, protonema
of mosses.

Sporulation

↓
formation of
asexual spores.
Ex. Algae, Fungi

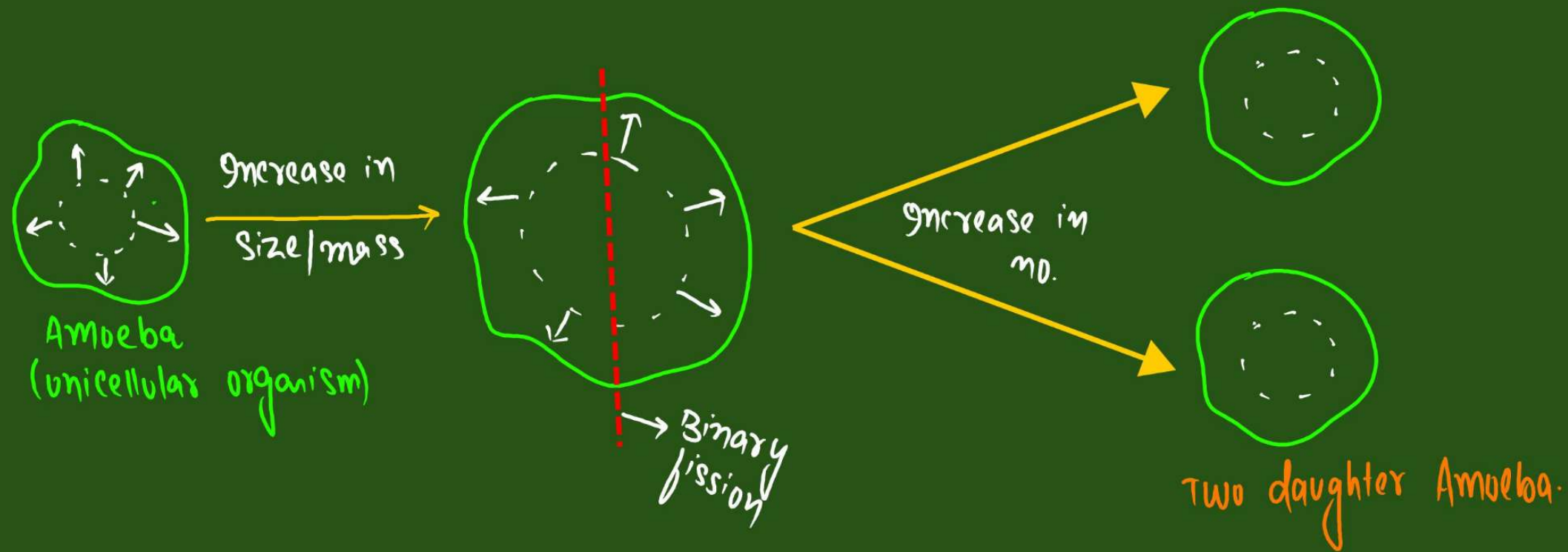
TRUE regeneration.

Ex. planaria
(Flat worm).

↳ Reproduction is absent in non-livings.

imp. ↳ Growth and reproduction is same/synonym — in case of unicellular organisms.

⇓
i.e, mutually inclusive events
(i.e, increase in no. of cells).



* Reproduction is not a defining property.

Because there are some living organisms in which reproduction is -nt.

Ex. Mule ————— Cross b/w ♂ donkey and ♀ horse
(Sterile)

Infertile human couples.

worker bees (honey bees).

(c) METABOLISM

↳ Sum total of all the chemical rxns. which occur inside living cell or living organism.

↳ Two types: a. Anabolism: Build-up rxns.

Ex. Photosynthesis



b. Catabolism: Break down rxns.

Ex. respiration

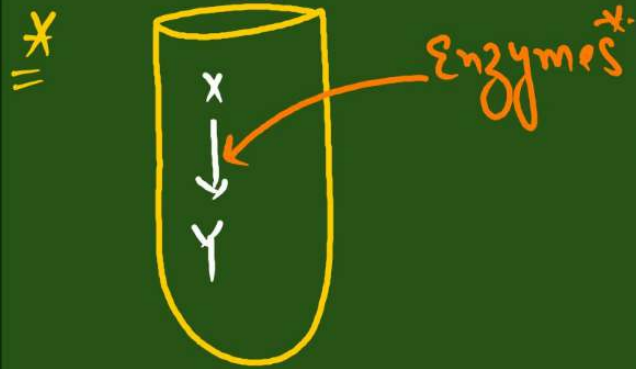


Note:- All the metabolic rxns are driven by biological enzymes which are +nt inside living cells

Q. Can a metabolic rxn. occurs in-vitro (i.e., outside living cell) like inside test tube?

Ans. $\left\{ \begin{array}{l} \text{a. Yes (Thumbs up)} (\checkmark) \\ \text{b. NO (Thumbs down)} \end{array} \right.$

metabolic rxn. can occur outside the living cell (in-vitro) but the enzymes required should be provided.

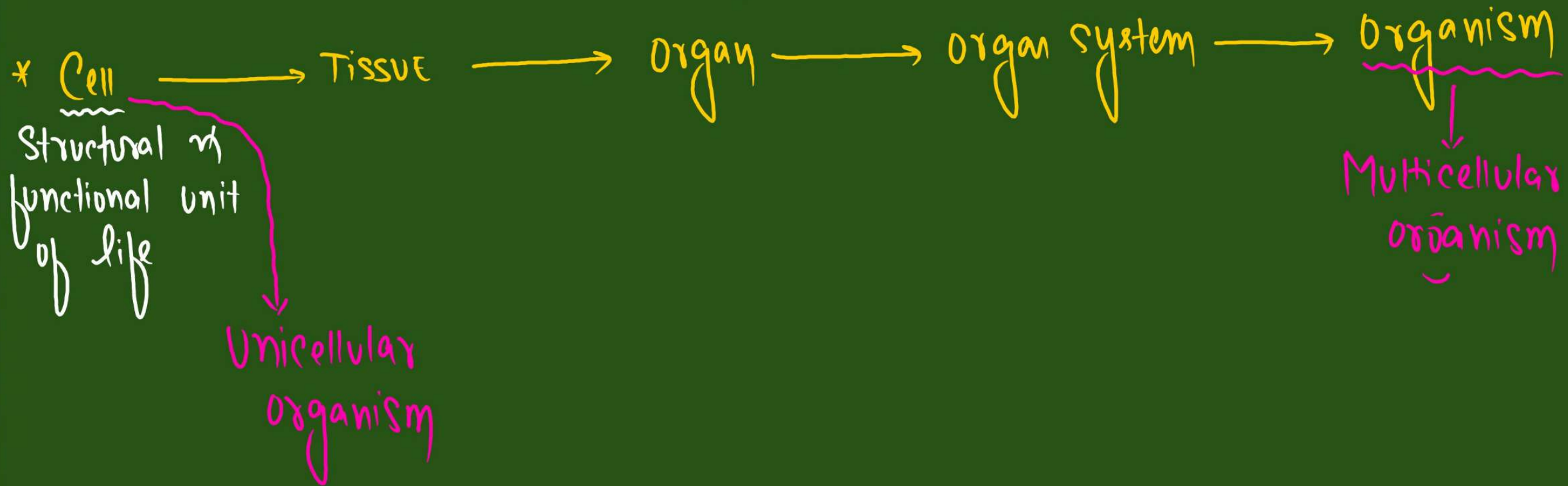


* Metabolic rxns. performed in-vitro are neither living organisms nor non-living. But surely it is a living rxn.

* Metabolism is a defining property because it is +nt in all living organisms without any exception.

(d) CELLULAR ORGANISATION

- ↳ All living organisms — are composed of cells.
- ↳ {
 - unicellular organism — single celled.
 - multicellular " — multiple cells.



⊗ Cellular organisation — is a defining property ($\because$ it is +nt in all living organisms without any exception).



(e) CONSCIOUSNESS

↳ Consciousness = Perception of Stimulus + Response.

≡ Stimulus can be

- biological (ex. mosquito bite).
- Chemical (ex. perfumes).
- physical (ex. temp.)

≡ Consciousness is a defining property.

* Ultimate level of consciousness is not in human beings

Kld as

"Self-consciousness"

↓
due to presence of well developed nervous system.

↓
(humans are aware of himself i.e, can plan future, think of past)

Note:- Self-consciousness is not a defining property because it is not only in humans

↳ It is difficult to define living state in humans.

Ex. Coma patients
(medical state)

Brain dead*

Virtually supported by machines.

possess consciousness (∵ they respond to drugs/chemicals entering into their body).

No self consciousness (∵ no coordination b/w different body parts).

Note :-

↳ All living organisms — Share same genetic material (i.e., DNA)
But to varying degrees.

QUESTION

H/w



Which one of the following aspects is an exclusive characteristic of living things?
[MR* 2011 Mains]

- 1 Perception of events happening in the environment and their memory
- 2 Increase in mass by accumulation of material both on surface as well as internally
- 3 Isolated metabolic reactions occur *in vitro*
- 4 Increase in mass from inside only

QUESTION

H/W

The living organisms can be unexceptionally distinguished from the non-living things on the basis of their ability for:

- 1 Photosynthesis
- 2 Metabolism
- 3 Growth and movement
- 4 Reproduction

QUESTION

H/W

All living organisms are linked to one another because:

- 1** The have common genetic material of the same type
- 2** They share common genetic material but to varying degrees
- 3** All have common cellular organization
- 4** All of above

Thank
You